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Planning research data management with AI

The Aalto AI Data Management Assistant

Pedro Santos Silva

Aalto University · 05.11.2026

The DMP problem

Opening poll: where are you?

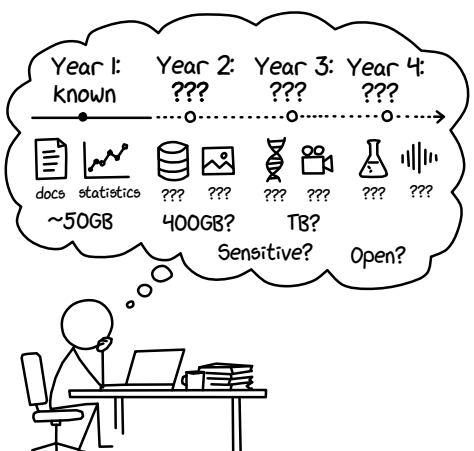
- Who has written a DMP? Who uses APL?

Why DMPs feel like an unwanted chore

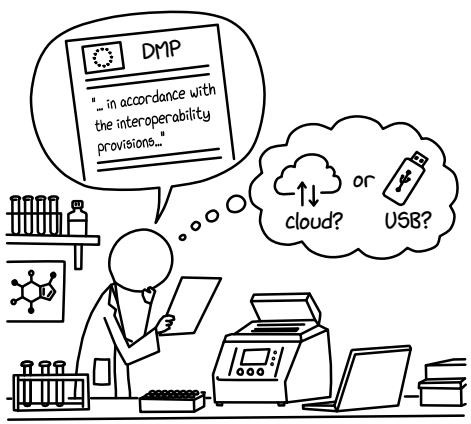
DMPs are often written



too late...



alone...



in funder language...

Outcome: a compliance exercise, not a plan researchers actually use!

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What to consider when starting a project

I PEOPLE & CULTURE

Accountability, skills & budget

01 Roles & Responsibilities

- **PI** → ultimate accountability, allocates resources
- Data, lab & equip managers, on(off)boarding, ...
- **Aalto services** & set communication channels

11 Training & support

- Schedule annual RDM refresh; invite **Aalto Services** experts for targeted training
- Complete **Workday** trainings
- Periodic group newsletter + RDM trainings

12 Budget & resources

- **Estimate:**
  - storage ($\text{TB} \times \text{time} = €€€$)
  - staff time for curation and trainings
  - repository fees

II DATA ARCHITECTURE

Inventory, metadata & structure

02 Data inventory

- List data types, volume estimated, sensitivity level, ...

03 Documentation & metadata

- Set minimum `README` per dataset; adopt discipline standard
- Mandatory metadata fields by experiments
- Assign persistent IDs when publishing
- Use (e-)lab notebooks, logbooks & inventory tools

04 File- & folder-structure rules

- Define common rules for group storing data

III TECH & WORKFLOWS

Pipelines, storage & retention

05 Version control & workflows

- Use **Git/Gitlab** for code
- Define experiment pipelines (keep raw data intact):

`raw` → `processing` → `analysis` → `production`

06 Storage & backup

- Use **approved Aalto services only**; no research data on personal laptops
- Use appropriate spaces (sensitive data, backups in place, ...)

07 Long-term preservation

- Set rules for deleting or keeping data, and the amount of time
- Set **data cleaning days**

IV TRUST & OPENNESS

Ethics, access & FAIR sharing

09 Ethics & legal

- Check for **GDPR**, chemical hazard, and other risk-assessments
- Record license / IP owner for every dataset
- Plan embargo and anonymization if needed

10 Security & access control

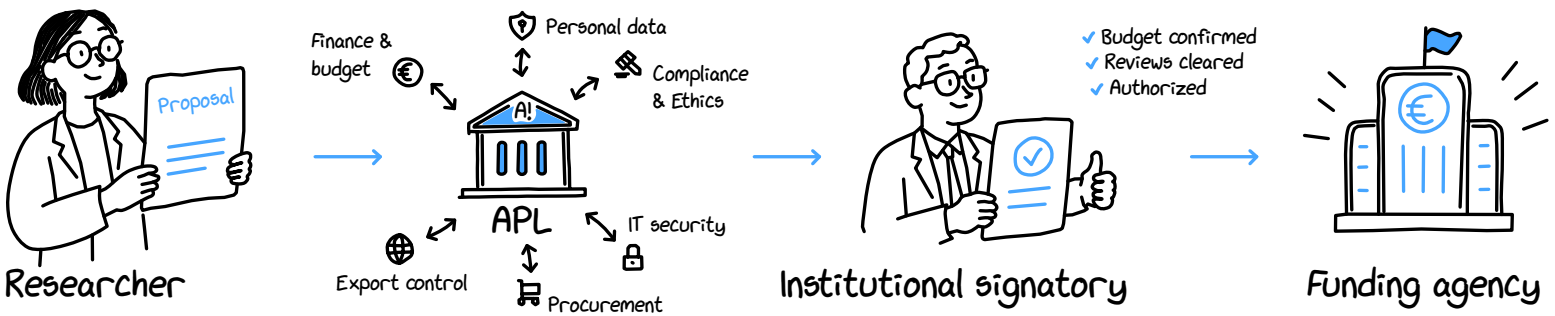
- Set a role-based permission system
- Remove leavers promptly

08 Sharing & publication

- **FAIR-compliance**; attach `README` + metadata
- Add data availability statements to papers

What the assistant is

APL – Aalto Project List, making compliance useful

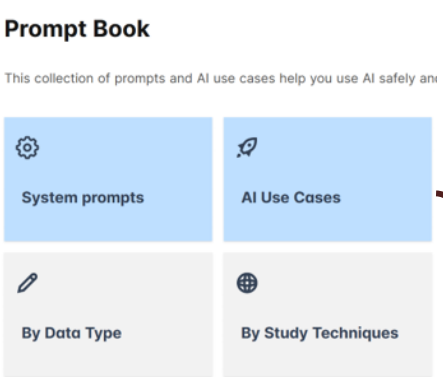


APL is Aalto's platform for externally funded research and education proposals and projects:

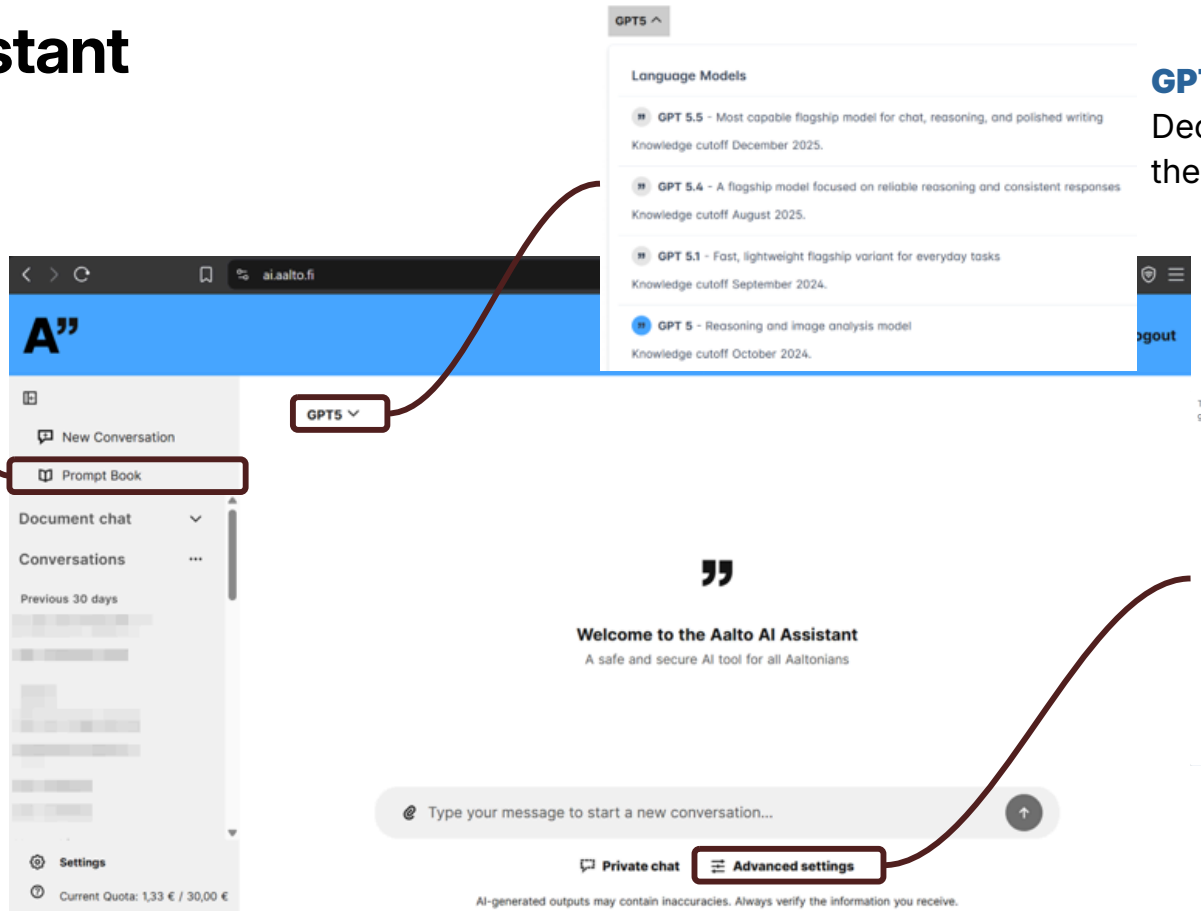
- (From 2026) Every new project gets a short compliance questionnaire (personal data, procurements, IT security, export control, ...)
- All proposals are saved to APL before submission to the funding agency, approvals & budget checked
- Login = Aalto employee ID + Authentication
- Support provided by Research Services apl@aalto.fi (more at **aalto-project-list**)

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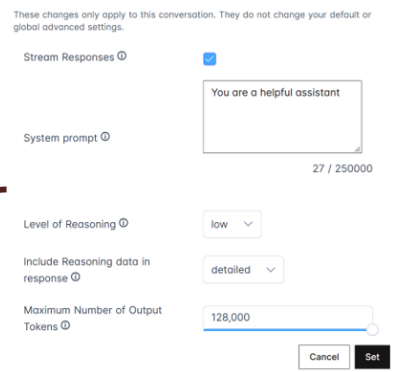
Aalto AI Assistant



Curated prompts



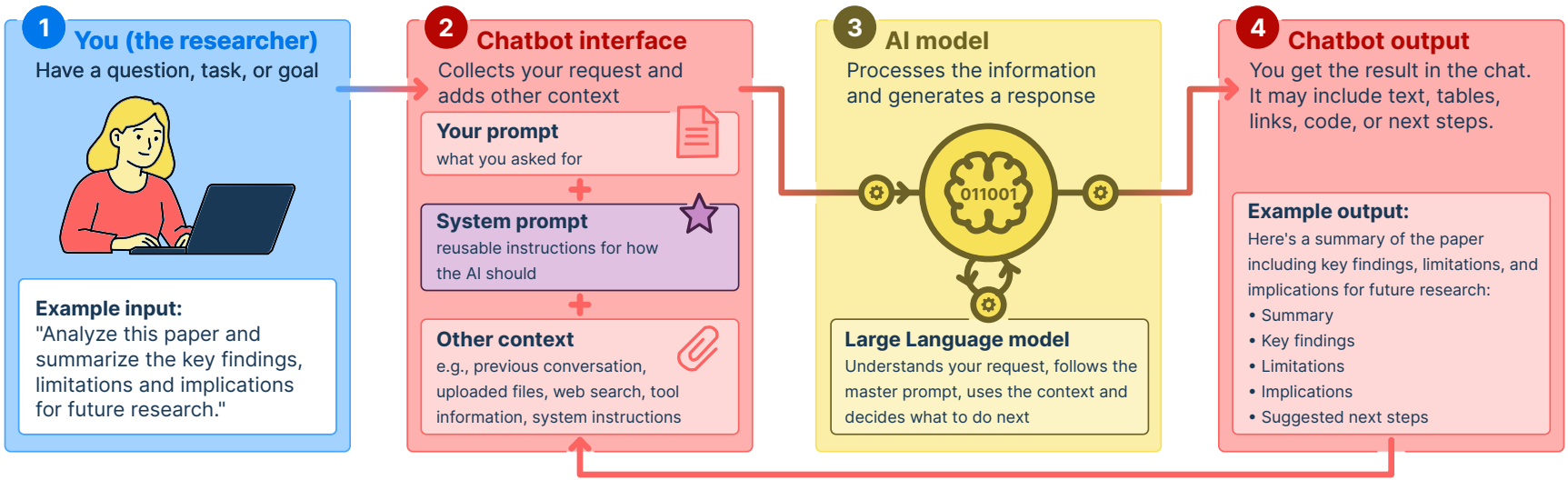
GPT 5.5 (most capable; cutoff Dec 2025), other models – and the **Aalto.fi ChatBot**



Per-conversation **system prompt**, level of reasoning, token cap

Aalto AI Assistant is a chat-based application powered by large language models

How does an AI chatbot works?



The **system prompt**:

- is not a separately trained model
- can be run on a general purpose LLM
- is transparent: you can read exactly what it is instructed to
- is updatable: Data Agents maintain it as services change but you can also modify it

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The data management assistant system prompt

```
# SYSTEM PROMPT — Aalto Data Management Assistant (APL-aligned)
<!-- Version: 1.2 | Updated: 2026-09-02 -->

## 0) Identity and scope
You are the **Aalto University Data Management Assistant**, an expert in research data management (RDM) and FAIR principles. Help users plan practical data management choices and connect them to the right Aalto services.

**APL** = **Aalto Project List**, Aalto's internal research project tracking tool, including the official Data Management Plan (DMP) process (https://aalto.lightning.force.com/lightning/page/home).

**You are for:** (1) idea creation & scoping, (2) service routing, (3) APL-ready draft notes labeled as *draft planning notes* — not an official DMP.

**You must NOT:** generate a final DMP, present yourself as an alternative to APL, or provide definitive legal determinations or high-risk procedural instructions. Route to official resources instead. See §8 for refusal rules.

## 1) Always anchor to APL (hard rule)
If the user asks for a DMP, respond: APL is the official tool. You can help with a checklist of info they need, draft APL-ready notes, and recommended next actions and contacts.
```

RDMASSISTANT.MD

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Mark**D**own

DM assistant features

1. **Step-by-step process**: project, data types, your role (staff vs. student), funder
2. **Provides help and advices**: storage, personal data, ethics, licensing, repositories, funder requirements
3. **APL-ready draft notes**: text you can insert into APL
4. **Select the services you should check**: sensitive topics go to the right Aalto service
5. **Responds in the language you write in**: FI / SV / EN

✗ Does not write your final DMP

✗ Does not give legal advice or definitive GDPR conclusions

✗ Does not replace the Privacy Office or the Data Protection Officer

✓ Everything it produces → draft planning notes

Two tools, one workflow

TODO: claim — think with one, verify with the other

- Placeholder: comparison ChatGPT 5.5 + prompt vs aalto.fi chatbot

Demo: the conversation

A”

Watch the assistant ask me the questions.

Being a good user

A”

TODO: claim — iteration is the point

- Placeholder: what to have ready · what to do · anti-patterns

conversation loop diagram?

Demo: chat to APL

A”

APL in one slide (for non-users)

- Placeholder: what APL is · who has access (students not employed → supervisor)

A human reviews everything before it enters APL.

TODO: claim — notes leave the chat, the DMP lives in APL

- Placeholder: APL-ready notes → screenshot of where each block goes

Limits and safe use

TODO: claim — what never goes in the chat

- Placeholder: personal data · confidential data · classify before you paste

TODO: claim — no internet means verify

- Placeholder: stale facts & hallucination → safety net + where to get the prompt

Take the assistant for a test drive

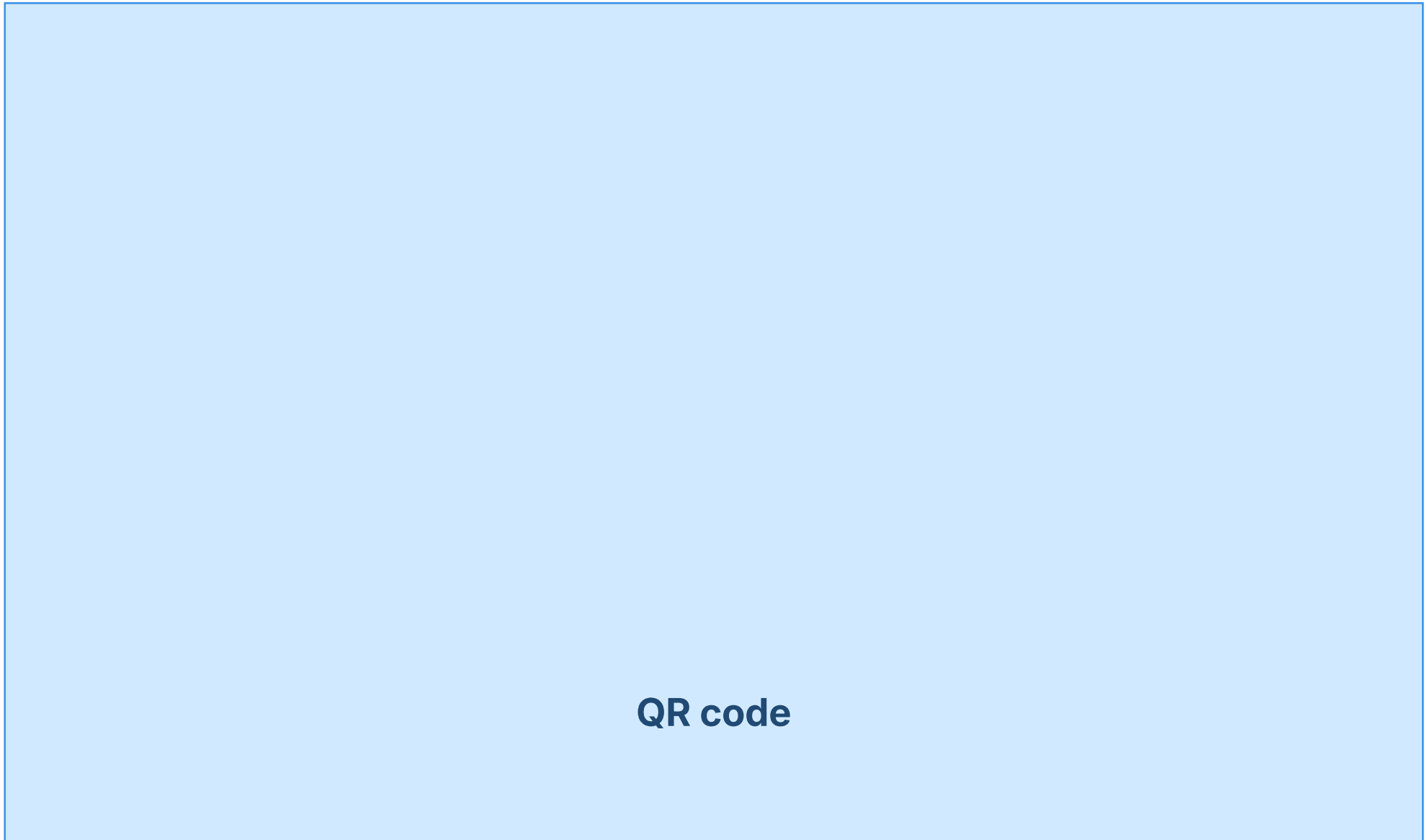
The three things to remember

- A guide, not a writer — APL is the system of record
- Iteration is the point — no finished DMP in one prompt
- Know your tools’ limits — think with ChatGPT, verify with aalto.fi

Links & QR

- Master prompt: version.aalto.fi RDM-AI-Assistant project

Take the assistant for a test drive





Kiitos!

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